PRODUCT & COMMERCIAL STRATEGY

Phone Loan & Inventory Management System

Reality Check, Capability Gaps & Monetisation Strategy — Tanzania Market

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1. Executive Summary

The existing technical specification is a sound transactional core. It describes, accurately and in useful detail, how a phone moves from a supplier box to a branch shelf, how a sale or loan is recorded, how stock is audited, and how an offline-first client should behave. As an engineering document it holds up. As a description of the business it is incomplete.

The central finding of this review is simple: **the specification models a stock system for a business whose money and risk actually sit in credit**. Phone retail in Tanzania is not primarily a margin business on hardware. Margins on the handset itself are thin and competitive. The durable profit centres are the financing of the handset, the collection of instalments, and the reduction of default losses. The current specification treats the loan as a form of sale. It should treat credit as the product.

Seven capability layers are missing. Each one is commercially material. Ranked by how much value they unlock or protect:

- **Gap 1 — Device locking and recovery.** Without a recovery mechanism, an unpaid device is unrecoverable and the loss is total. This is the single largest lever on default loss.
- **Gap 2 — Automated mobile money repayment rails.** Cash collection at the counter is slow, manually reconciled and easy to falsify. Programmatic collection changes both cost and control.
- **Gap 3 — Borrower identity, KYC and guarantor mapping.** A credit product without verified identity cannot price risk, cannot enforce, and cannot scale responsibly.
- **Gap 4 — Agent and consignment inventory.** A large share of real-world unit movement happens through third parties. A ledger that only understands branches under-counts reality.
- **Gap 5 — Dual-IMEI trapping, trade-in and blacklist registry.** Treating a dual-SIM handset as one identifier leaves a bypass path open on every locked device.
- **Gap 6 — Fiscal compliance and receipting.** Sales in Tanzania operate inside a tax receipting regime. Ignoring it limits which customers can legally adopt the system.
- **Gap 7 — Credit decisioning and collections operations.** Scoring, arrears ladders and restructuring are workflows, not reports. They need to exist as first-class states in the system.

Each gap is described in Section 6 with the risk of omitting it, an estimate of build complexity, and its commercial leverage.

Section 8 sets out a monetisation architecture built on six streams: subscription, per-device loan facilitation, payment rail margin, messaging margin, white-label deployment, and guarded risk analytics. Section 9 works an illustrative unit-economics model. At a blended revenue of roughly TZS 440,000 per active shop per month, an installed base of about 260 active shops implies annual revenue in the region of TZS 1.37 billion by year three. These figures are planning assumptions, not forecasts, and every input is listed so it can be challenged.

> **THE ONE SENTENCE VERSION**
> Finish the transactional core as specified, then immediately extend the schema so that **credit, collection and recovery** are native to the data model. Sell the system as a control tool for a lending business, not as a stock register.

2. Purpose, Scope & How to Read This Document

This is the companion to the technical blueprint (document LIMS-SDD-2026-001). That document answers **how the system is built**. This one answers two different questions: **is the specification complete enough to run a real Tanzanian phone dealership**, and **how does the resulting product earn money**.

2.1 Audience

- **Product and engineering** — Sections 5, 6 and 13 define what must be added to the specification and where it attaches in the data model.
- **Commercial owners** — Sections 8, 9, 10 and 14 define pricing, packaging, go-to-market and the sequence for switching revenue on.
- **Both** — Section 11, the legal and regulatory guardrails, and Section 12, the risk register. These are shared constraints, not engineering details.

2.2 What this document is not

- It is not a replacement for the technical blueprint. Where the two overlap, the blueprint is authoritative on implementation detail.
- It is not a forecast. Every currency figure is a planning assumption chosen to make the arithmetic visible and challengeable.
- It is not legal advice. Section 11 identifies the areas that require qualified local legal review before launch; it does not substitute for it.

2.3 How confidence is labelled

This review draws on three different kinds of evidence, and they do not carry equal weight. Where a statement comes from direct inspection of the codebase it is marked **Observed in codebase**. Where it describes market conditions in Tanzania it is marked **Market observation**. Where it is an estimate or a planning input it is marked **Assumption to validate**. The distinction matters: the first category can be relied on now, the third cannot.

3. Assessment Method & Evidence Base

The review was conducted in two passes.

- **Pass one — codebase inspection.** The application directory was read directly. The Flutter client contains 37 Dart source files at approximately 6,430 lines of hand-written application code, a complete navigation graph, a localisation layer and 16 reusable design-system widgets. The data layer is a 12-table PostgreSQL schema with foreign keys and indexes. The service layer returns mock data; the live Supabase calls are present but commented out, and the backend client still carries placeholder credentials.
- **Pass two — specification review and market framing.** The 20-requirement specification was read against the observed codebase, then against the operational reality of phone retail and handset financing in Tanzania: multi-branch dealers, Kariakoo wholesalers, regional shops, third-party sellers, dual-SIM handsets, mobile money dominance and fiscal receipting obligations.

Three constraints applied throughout: the review does not assume access to proprietary competitor systems; it treats currency figures only as illustrative; and it flags anything requiring legal or regulatory confirmation rather than resolving it by assertion.

Evidence class	What it covers	Reliance
Observed in codebase	File counts, line counts, schema tables, mock service layer, placeholder credentials, dead navigation targets	High — reproducible
Market observation	Structure of Tanzanian phone retail, credit practice, mobile money usage, receipting norms	Medium — directional
Assumption to validate	All pricing, adoption rates, default assumptions, unit economics	Low — must be tested

4. Reality Check — What the Specification Already Gets Right

Before listing what is missing, it is worth being precise about what is strong. The following choices are correct and should not be revisited.

- **Separation of the branch interface from the management layer.** The instruction that an employee should not see an enterprise dashboard is a genuine product insight, not a simplification. Counter staff operate under time pressure and poor lighting with a customer waiting. Complexity on that screen becomes data quality problems in the system.
- **The phone, not the IMEI, is the entity.** Treating the IMEI as an identifier and the device as the persistent record is the correct modelling choice. It leaves room for identity changes, refurbishment and dual identifiers without rewriting the core.
- **Stock derived from movements rather than an editable quantity.** This is the difference between a register and a ledger. It is what makes it possible to answer why a branch has a particular count. The specification states this explicitly and it is right.
- **Explicit transfer states.** Draft, approved, in transit, received, closed. Ownership is unambiguous at every moment, which is exactly what matters when a dispute arises between two branches.
- **Queue-then-validate offline behaviour.** Refusing to let offline mode finalise conflicting operations blindly, and rejecting last-write-wins for stock ownership, is a mature decision. Most first-generation attempts at this fail here.
- **AI layered above validated data with no write access.** Stating that the intelligence layer must not become the source of truth, and must hand sensitive actions back to authorised workflows, prevents the most common failure mode of AI features in operational software.
- **Phasing that puts AI last.** Sequencing the assistant after the transaction and inventory model is reliable is correct, and rarely followed.

> **VERDICT**
> The transactional core is well specified. Nothing in Sections 5 and 6 argues for rebuilding it. The critique is about **completeness**, not correctness: the specification describes the system accurately, but it describes only part of the business.

5. Reality Check — Where the Specification Falls Short

Five structural weaknesses run through the document. They are not oversights in individual requirements; they are gaps in the frame the requirements are written in.

5.1 It models a stock system for a credit business

The specification treats a loan as a variant of a sale: scan the device, choose sale or loan, capture customer and terms, confirm. That is the correct sequence for a sale. It is a thin description of a loan. A loan is not a transaction that completes; it is a relationship that begins, with a schedule, an expected cash curve, a collection process and a set of failure states. None of those appear anywhere in the specification.

Consequence: the system will record that a loan was issued, and will have no native way to say whether it is healthy, in arrears, restructured, written off or recovered. **Market observation** — driven by the phone itself: for a dealership, the profit from a financed handset is realised over weeks or months, and so is the loss.

5.2 Collections are absent as a designed workflow

The specification requires that loan balances and schedules be calculated from controlled terms, and that manual balance edits be restricted and audited. That is the accounting requirement. It is not the operational one. There is no mention of reminder timing, no arrears ladder, no escalation path, no record of what was said to a customer, no tracking of partial payments beyond the balance arithmetic, and no mechanism to act on a missed instalment.

Consequence: collection becomes something staff do outside the system, in their heads or in a notebook. The system then holds an incomplete picture of exactly the thing the business most needs to control.

5.3 Borrower identity is under-specified for a lending product

Identity appears in the specification as a customer relation: a customer identifier, a sale identifier, a loan identifier. Phone number, name and a photograph are implicit at best. For cash sales that is adequate. For credit it is the risk model. Nothing in the specification addresses the difference between a verified person and an entered name.

Consequence: the system cannot support responsible credit decisions, cannot support enforcement, and cannot distinguish a first-time borrower from someone with an unpaid balance at another branch of the same dealership.

5.4 Offline-first is specified for the app, not for the counter

The offline architecture is technically well reasoned: a local database, an outbox, a client operation identifier per transaction, a sync worker, server-side validation on arrival, and explicit conflict reconciliation. What is missing is the operational design. What does the counter display when a transaction is queued but unconfirmed? What can staff promise a customer in that state? What happens at daily closing when a queued transaction has not yet been acknowledged by the server, and therefore appears in the local count but not the central one? What is the procedure when two devices conflict on the same handset, and who resolves it?

Consequence: the hard part of offline-first in field conditions is not synchronisation. It is the human protocol around an unconfirmed state. Undefined protocols get resolved informally, and informal resolution is unauditable.

5.5 No fiscal or statutory layer

The systems this product replaces are not purely internal. Sales in Tanzania sit inside a fiscal receipting regime, and businesses above registration thresholds are expected to issue compliant receipts through approved electronic fiscal devices. The specification describes reporting as an internal management tool and does not address receipting, tax

records or statutory reporting.

Consequence: larger and more formal dealerships — exactly the customers with the volume and capital to pay for a system like this — cannot adopt it as their system of record without a parallel process. That is an adoption ceiling written into the design.

5.6 Nothing converts the handset into a recovery asset

The specification is strong on custody and ownership. It can tell you which branch holds a device and which customer received it. It contains no concept of the device as collateral, and therefore no mechanism that gives a financed handset any recoverable value after default.

Consequence: a default is a total loss rather than a partial recovery. In an unsecured credit business, this single design omission typically outweighs every other risk-control measure combined. It is treated in detail as Gap 1 in Section 6.

Weakness	Commercial consequence	Severity
Loan modelled as a sale variant	Credit health cannot be seen or managed natively	High
Collections not a workflow	Staff manage debt outside the system; poor oversight	High
Identity under-specified	No risk pricing, no enforcement, no borrower history	High
Offline protocol undefined for staff	Informal workarounds; unauditable exceptions	Medium
No fiscal/statutory layer	Formal, higher-volume dealerships cannot adopt	Medium
No recovery asset concept	Default equals total loss instead of partial recovery	Critical

6. Capability Gap Register

Seven capability layers are recommended for addition. Each is described below with the risk of omitting it, an indication of build complexity, and the commercial leverage it carries. Complexity is expressed in relative terms (low, medium, high) because absolute effort depends on the team's familiarity with the stack.

Ref	Capability	Risk if omitted	Complexity	Revenue leverage
G1	Device locking & recovery	Default is a total loss; no recovery path	High	Protects core margin
G2	Automated mobile money repayments	Slow collection, manual reconciliation, fraud exposure	Medium	Enables rail margin
G3	Borrower identity, KYC, guarantors	Cannot price risk or enforce; no borrower history	Medium	Enables credit at scale
G4	Agent & consignment inventory	Ledger under-counts real unit movement	Medium	Extends addressable market
G5	Dual-IMEI trapping, trade-in, blacklist	Bypass path on every locked device; stolen stock re-enters	Medium	Protects G1 effectiveness
G6	Fiscal compliance & receipting	Formal dealerships cannot adopt as system of record	Medium	Unlocks upper segment
G7	Credit decisioning & collections ops	Arrears handled informally; no escalation discipline	High	Reduces loss, enables scale

6.1 G1 — Device Locking & Recovery

This is the highest-leverage gap. In an unsecured handset financing business, the financed device is the only asset securing the loan. If defaulting on that loan has no consequence for the customer's ability to keep using the handset, the expected loss on every loan is materially higher, and the business compensates by lending less, charging more, or refusing applications. All three outcomes shrink the market.

There are two credible implementation paths, and the distinction matters commercially and legally.

- **OEM-backed financing lock (recommended for v1).** Established device-finance lock programmes operate at the operating-system level with the manufacturer's cooperation: the device is provisioned at point of sale as a financed device, and the lock state is enforced by the platform. These programmes are commercially sanctioned, contractually governed, and legally defensible. They require a commercial agreement, per-device participation, and integration work, but they are the supportable route.
- **Custom device-owner provisioning.** Building a bespoke kiosk or device-owner application requires provisioning privileges on the handset, cooperation at activation, and a robust remote command path. It is technically achievable but far more fragile: it depends on provisioning surviving user attempts to remove it, on the device remaining online, and on the specifics of the Android version and vendor skin. It also carries materially higher legal exposure if not disclosed and contracted properly.

Whichever path is chosen, four system elements are required and none of them exist today:

- A **lock state machine** on the device record: unlocked, financed, grace period, locked, unlocked-by-payment, recovery-initiated, recovered, written off.
- A **lock event ledger** that records every state transition with reason, actor and server timestamp. Locking a customer's device without a traceable authorisation record is an operational and legal hazard.

- A **clear repayment covenant** appearing in the customer agreement: the conditions under which a device may be locked, the notice given, the amount required to release it, and the mechanism for dispute.
- A **release path that is instant and automatic** the moment payment is confirmed, with no human intervention required. The commercial credibility of the whole mechanism depends on the release being faster and more reliable than the lock.

> **GUARDRAIL**
> Locking is only defensible when it is **disclosed, contracted, proportional and reversible without friction on payment**. Section 11 sets out the legal review that must precede any deployment. A covert or undisclosed capability should not be built: it converts a legitimate credit control into a legal liability and destroys the trust the product depends on.

6.2 G2 — Automated Mobile Money Repayment Rails

Manual cash collection at the counter is the default in most Tanzanian phone dealerships, and it is expensive in ways the specification does not capture: staff time, physical cash handling, manual reconciliation against a ledger, and a persistent exposure to falsified payment claims. Mobile money has the opposite profile: the transaction is already digital, already timestamped, already identified to a payer, and already reconcilable programmatically.

The system should support three integrated collection mechanisms, in ascending order of automation:

- **Push prompt collection.** On the due date, the system prompts the customer on their handset to authorise a payment to the dealership. Authorisation requires only the customer's mobile money PIN. This is the primary mechanism because consent is explicit and per-transaction.
- **Standing arrangement with the customer.** Where the customer and the operator support it, a scheduled collection agreement can reduce per-instalment friction further, still subject to the customer's authorisation framework.
- **Merchant collection account with webhook reconciliation.** A business collection account with an aggregator delivers payment events to a server endpoint. Each event is matched to a loan schedule entry by payer reference and amount, the payment record is created, the balance is recomputed, and any lock release is triggered automatically.

The design requirements are specific: every payment event must be matched idempotently, so a duplicate webhook cannot double-credit a customer; unmatched events must land in a reconciliation queue rather than being silently dropped or guessed at; and partial payments must be represented explicitly rather than rounded into the schedule.

Market observation — the mobile money ecosystem in Tanzania is fragmented across operators, which is why integration should be pursued through an aggregator rather than directly with each operator. The aggregator layer also normalises webhook behaviour and provides a single settlement relationship.

6.3 G3 — Borrower Identity, KYC & Guarantor Mapping

Credit requires identity. The specification currently treats the customer as a record attached to a sale. For a lending product the customer record must carry enough verified identity to support three separate functions: a credit decision, an enforceable agreement, and a recovery path when the agreement is breached.

- **National identification.** Tanzanian national identity verification through the issuing authority is not a freely available public interface. Commercial access generally requires a formal agreement. The practical v1 approach is to capture and store identity document details, record a clear photograph of the document, and record a live photograph of the borrower at the point of issuing the loan, so that identity is at least documented and comparable. Programme the authority interface as a later integration once an agreement is in place.

- **Verified mobile number.** The mobile money account used for repayment should be linked to the identity captured. A repayment rail that accepts money from an unrelated payer reference undermines the matching logic and the identity record simultaneously.
- **Guarantor mapping.** Co-signing is standard practice in this market and is a strong loss mitigant, but it only works if the guarantor is recorded with the same rigour as the borrower: verified identity, verified contact, explicit recorded consent, and visibility of the obligation they are accepting.
- **Consent and purpose limitation.** Personal data captured for credit assessment must be recorded with its purpose and the consent under which it was collected. This is both a regulatory requirement and a practical prerequisite for the analytics described in Gap 7.

One additional function is easy to miss and disproportionately valuable: **cross-branch borrower history**. If the same person has an unpaid balance at one branch, every other branch of the same dealership should see that at the point of application. This is a pure software capability and requires no external integration at all.

6.4 G4 — Agent & Consignment Inventory

The specification assumes devices move between branches and are sold at branch counters. In practice a significant proportion of unit movement happens through third parties: street sellers who take stock on consignment and settle after sale, small satellite shops supplied on credit, and sales agents working on commission. A ledger that recognises only branches will be systematically wrong about where devices actually are.

Required additions to the model:

- An **agent record** distinct from both employee and customer, with its own identity, contact and settlement terms.
- A **custody state** for devices held by an agent, distinct from in-transit and distinct from sold. The device has left the branch but is not sold; ownership and liability sit with the agent.
- A **consignment agreement** capturing the devices issued, the agreed commission or transfer price, the settlement deadline, and the return expectation.
- A **settlement workflow** that resolves the consignment at the end of its period: devices sold, devices returned, devices outstanding, cash due. Outstanding devices on an expired consignment should raise an alert and appear in the discrepancy register.
- **Agent performance visibility** — sales rate, settlement punctuality, return rate, loss rate. This is the raw material for deciding who is worth extending more stock to, and it is only obtainable if agent custody is modelled properly in the first place.

6.5 G5 — Dual-IMEI Trapping, Trade-In & Blacklist Registry

The specification treats IMEI as an identifier and requires uniqueness enforcement at the database layer. That is correct but incomplete for two reasons.

First, most handsets sold in this market are dual-SIM and carry two IMEI values. If only one is recorded and enforced, the second slot remains a bypass: a locked or blocked device can be brought back into service by changing which slot holds a SIM. Both identifiers must be captured at intake, both must be constrained for uniqueness, and the capture step should read both in a single scan of the box barcode so that staff are not asked to perform a second, easily-skipped action. The mobiles being handled here are dual-SIM consumer handsets; there is no remotely modifiable device state involved, only the pairing of the two identifier values to one device record.

Second, there is no concept of provenance. The system knows what a device is, not where it came from. Trade-in is a normal part of this market — a customer brings in a used handset as part payment, or a shop buys stock from a walk-in

seller. Without a check at intake, a stolen device is accepted into inventory in good faith, at which point the dealership holds a liability and has lost the purchase price.

The recommended addition is a **blacklist registry with intake verification**: a table of flagged identifiers carrying a reason and a source, checked automatically against every intake operation, whether the intake is a supplier batch, a trade-in or a walk-in purchase. The registry is only useful if the check is automatic and blocking rather than advisory, and if a flagged match produces a clear explanation rather than a generic error.

Trade-in valuation is a natural companion capability: a small valuation table keyed by brand, model, storage and condition, so that trade-in offers are consistent across branches and auditable rather than negotiated per counter.

6.6 G6 — Fiscal Compliance & Receipting

This gap is commercial rather than operational. The specification positions reporting as an internal management tool, which is correct as far as it goes, but it means the product cannot serve as the system of record for a dealership operating within the formal tax framework.

- **Electronic fiscal receipting.** Tanzanian businesses operating above the registration threshold are expected to issue compliant electronic fiscal receipts through approved devices or systems. A sale recorded only in an internal ledger does not satisfy that obligation.
- **Tax-relevant transaction fields.** Compliant receipting requires that each sale carries the fields a fiscal receipt needs, at the moment it is created, not reconstructed later from a report.
- **Statutory reporting.** Monthly and annual obligations draw on the same transaction data. If the data model is designed with those fields present from the start, statutory reporting becomes an export; if not, it becomes a reconciliation project every filing period.
- **Sequencing and format constraints.** Fiscal receipting regimes typically impose requirements on receipt numbering, formats and submission that differ from internal record-keeping conventions. This must be confirmed with qualified local advisers and then designed in, not retrofitted.

Assumption to validate: the specific integration route (certified device, certified software, or intermediary service) and the registration thresholds that trigger the obligation in the dealership segment being targeted. This requires confirmation from a qualified Tanzanian tax adviser before it is treated as a design input.

6.7 G7 — Credit Decisioning & Collections Operations

This is the gap that determines whether the business scales safely. Gap 3 provides identity; Gap 7 uses it. Without a decision layer, credit decisions are made per counter by individual staff, which means the loss rate is a function of staff judgement rather than of policy, and it is invisible until it is expensive.

- **A simple, explicit scoring rule.** At v1 this does not require a statistical model. A transparent score built from a handful of observable factors — verified identity, verified mobile money account with transaction history, a guarantor on record, down payment proportion, prior repayment record within the dealership, tenure of employment or business — is enough to produce consistent decisions across branches. Transparency also means the rule can be explained to a customer and defended if challenged.
- **Affordability parameters at the point of sale.** A structured terms calculator that derives instalment, duration, deposit and total obligation from the device price, so that the counter is not negotiating arithmetic.
- **An arrears ladder as machine states.** Not a report: a defined sequence of states a loan passes through as it becomes overdue, each with a trigger day, an action, an owner and an evidence requirement. Reminder, escalation, guarantor contact, lock, formal recovery, restructure, write-off. Every transition recorded.

- **A collections action log.** What was done, when, by whom, and what the customer said. This is what turns collections from an informal activity into an auditable one, and it is the record that supports enforcement if it becomes necessary.
- **Restructuring as a first-class operation.** Payment holidays and rescheduled terms are normal and often better than default, but only if they are recorded as a formal change with a reason and an approver. Restructures that live outside the system destroy the integrity of every portfolio metric.

6.8 Secondary Gaps

The following are smaller in scope but should be recorded so they are not discovered late. They are not prerequisites for launch.

- **Accessory and voucher inventory.** A phone sale usually carries a case, screen protector and sometimes top-up credit. These are stock items with their own margins and their own shrinkage. A system that tracks only handsets tracks perhaps two-thirds of the retail business.
- **Warranty and repair intake.** Devices returned for repair create a custody state that is neither available-for-sale nor sold. Without it, repaired devices are either invisible or wrongly counted as available.
- **Swahili-first interface and low-literacy safety.** The specification mentions localisation. It should go further: the branch interface should be designed to be operable with minimal reading, using iconography, colour and scan-first interaction, because a meaningful share of counter staff will work in that mode.
- **Aging stock and pricing intelligence.** A device that has sat for ninety days is quietly losing value. An aging register with configurable thresholds turns an invisible cost into a visible one, and is the natural first use of the reporting layer.
- **Returns, damage and write-off states.** These are named in the daily closing workflow but need their own modelled states and audit reasons, otherwise they become informal adjustments.

7. Out of Scope — Deliberate Non-Goals

Scope discipline is what keeps the delivery plan credible. The following are explicitly excluded, and the exclusion should be recorded rather than left to interpretation.

Excluded	Reason
Acting as a deposit-taking or licensed lending institution	Requires a regulatory structure and licence that software cannot substitute for. Where interest-bearing lending is required, it should be done in partnership with an appropriately licensed entity.
Direct national identity authority integration in v1	Commercial access generally requires a formal agreement. v1 documents identity; it programmes the interface later.
Covert or undisclosed device locking	Excluded as a matter of principle and of legal exposure. Locking is only within scope when disclosed, contracted and reversible on payment.
Full accounting or ERP replacement	The system is a transaction and inventory control system. General ledger, payroll and statutory accounts remain with specialist tools, connected by export.
Any AI write access to transactional tables	Retained from the technical specification: the intelligence layer reads validated data and hands sensitive actions back to authorised workflows.
Hardware manufacture or distribution	Procurement and bundling of scanners and devices is in scope as a commercial activity; designing or manufacturing hardware is not.
Physical security, CCTV, biometric terminals	Deliberately avoided as dependencies. Identity and attendance are built on authenticated sessions and server timestamps instead of specialised hardware.

8. Monetisation Strategy — Tanzania

The pricing philosophy follows one principle: **charge where value is created, and charge in proportion to the money the system protects or unlocks.** A stock register saves administrative effort, which is a weak basis for pricing and a weak reason for a shop to pay. A system that reduces default loss, speeds up collection and makes recovered value possible protects real money, and can be priced against that.

8.1 Revenue Architecture

Six streams, ranked by expected contribution at scale. Streams 3 to 5 are transaction-linked and therefore scale automatically with customer success, which makes them the most durable.

#	Stream	Basis	Indicative price	Notes
1	Subscription	Per branch, per month, tiered by capability	TZS 40,000 – 500,000	Predictable base; anchors the relationship
2	Loan facilitation fee	Per device financed through the system	TZS 2,000 – 5,000	Tied directly to lending volume
3	Payment rail margin	Per repayment collected through integrated rails	TZS 300 – 500 equivalent	Grows with repayment frequency
4	Messaging margin	Per reminder or notification delivered	TZS 25 – 35 charged, 12 – 15 cost	Highest gross margin stream
5	White-label deployment	Setup plus recurring hosting and maintenance	Setup TZS 3M – 7M + monthly	For wholesalers and distributors
6	Risk analytics (guarded)	Aggregate, consented, non-personal insight products	Partner-negotiated	Requires the strongest privacy framework; latest to activate

8.2 Subscription Tiers

Tiers are differentiated by capability, not by seat count, because the value is in control rather than in user numbers. Every tier includes the offline-first client, because that is a requirement of the market rather than a premium feature.

Tier	Target customer	Included	Monthly price (TZS)
Shop	Single branch, cash trade, small loan book	Inventory ledger, IMEI registration, sales, transfers, daily closing, attendance, basic reporting	40,000 – 60,000
Multi-branch	Two or more branches, active loan book	Everything in Shop, plus inter-branch control, consignment and agents, collections workflow, KYC and guarantors, alerts, standard reports	100,000 – 180,000
Wholesale / Distributor	Importers and distributors supplying dealers	Everything in Multi-branch, plus box manifest ingestion, container and batch tracking, blacklist registry, white-label option	300,000 – 500,000
Enterprise (white-label)	Distributors deploying under their own brand to a dealer network	Full platform under the distributor's identity, hosting, maintenance, revisions	Setup plus negotiated monthly

Pricing guardrail: do not charge per IMEI scanned, per device registered or per user seat. Every one of those meters taxes exactly the behaviour the system needs most — full registration of stock and full adoption by staff — and would create a direct incentive to under-record.

8.3 Loan Facilitation Fee

This is the most important stream commercially, because it converts customer lending volume directly into revenue without requiring the customer to make a separate purchase decision. A fee per financed device is defensible because the system is doing real work at that moment: identity capture, agreement generation, schedule calculation, lock provisioning, and the collection rails that follow.

The fee should be positioned as a cost of doing credit business, not as a software charge, and it should be charged only on financing that actually completes through the system. That makes it self-justifying: the shop pays it out of the margin on a loan it would not otherwise have been able to make safely.

Assumption to validate: the level at which a per-device fee remains attractive relative to the shop's per-loan margin. This must be tested with real dealerships before it is fixed in a contract.

8.4 Payment Rail Margin

Where repayments flow through the integrated collection rails, a per-transaction margin applies. This stream has a structural advantage: it grows with repayment frequency, which is high, because instalments are frequent and small. A single financed device may generate eight to twelve repayment transactions over its term, each carrying margin, spread across the customer relationship.

It also aligns incentives correctly. The more efficiently the system collects, the less the shop spends on manual reconciliation, and the more transactions pass through the rails. Both parties benefit from the same outcome.

8.5 Messaging Margin

Automated reminders before a due date, on the due date, and after a missed instalment have a direct effect on collection performance, and they run on infrastructure with a fixed cost per message and a charge per message. The margin is the difference, and at scale it compounds: message volume is proportional to the size of the loan book, not to the number of shops.

Guardrail: message volume should be capped per loan and per day. Message fatigue is real, and a reminder channel that customers learn to ignore destroys more value than the margin it generates.

8.6 White-Label & Hardware Bundling

Large wholesalers eventually want the platform under their own name for their dealer network. This is a high-value, low-volume stream: fewer customers, much larger contracts, and a strong retention profile once a distributor's dealer network is running on the platform. Setup is a substantial one-off engagement followed by recurring hosting and maintenance.

There is a natural companion sale: handheld Android terminals and Bluetooth barcode scanners. Shops that adopt scan-first intake get materially better data quality than shops that type, and the hardware bundle removes the last excuse. Sourcing hardware to resell at wholesale is a commercial activity that requires no engineering.

8.7 Risk Analytics — Deliberately Last

Aggregated, consented, non-personal insight about lending performance across a portfolio is genuinely valuable — to dealers benchmarking their own performance, and potentially to partners assessing risk. It is also the stream with the greatest potential to damage trust and to breach data protection obligations if handled carelessly.

It is therefore sequenced last and gated on three conditions: the underlying transaction data must be reliable, the consent framework must explicitly cover derived analytics, and no output may expose personal data or re-identifiable

borrower detail. If those conditions cannot be met, this stream should simply not be activated. It is optional; the other five are not.

8.8 What Not To Do

- **Do not meter the basics.** Per-scan, per-IMEI or per-user charging penalises the data completeness the whole system depends on.
- **Do not price the small shop out of the entry tier.** The single-branch segment is the volume base and the source of referrals. A tier that a Kariakoo counter cannot justify will not be bought, and the base never forms.
- **Do not bundle messaging into the subscription at v1.** It has real marginal cost per send. Bundling it creates a cost exposure that grows with customer success, which is the wrong direction.
- **Do not charge for the recovery capability as a premium feature.** Device locking and recovery protects the customer's capital. It belongs in the product, and its cost belongs in the facilitation fee.
- **Do not discount for a term longer than the pilot.** Long free periods in this market rarely convert; they train the market to expect free.

9. Unit Economics — Illustrative Model

The purpose of this section is not to forecast. It is to make the arithmetic visible so that each input can be argued with individually. Every figure below is an **Assumption to validate**. The model uses a single metric — revenue per active shop per month — and then applies it to an adoption curve.

> **READ THIS FIRST**
> Nothing in this section has been validated against a real dealership. Treat the output as a structure for discussion, not as a projection. The value of the model is that it identifies which four numbers the business actually depends on, and they are all testable within one pilot.

9.1 Assumptions

Input	Value used	Why this value	Confidence
Financed devices per shop per month	60	Mid-sized dealership with an active loan book; the volume driver for the facilitation fee	Assumption
Facilitation fee per financed device	TZS 3,000	Mid-point of the proposed TZS 2,000 – 5,000 band	Assumption
Repayment transactions per shop per month	500	Approximately eight instalments per loan, spread across an active book	Assumption
Net rail margin per repayment	TZS 350	Mid-point of the proposed TZS 300 – 500 margin band	Assumption
Messages per shop per month	700	Three reminders per active loan across a book of roughly 230 active loans	Assumption
Net messaging margin per message	TZS 15	Charged at TZS 25 – 35 against a cost of TZS 12 – 15	Assumption
Blended subscription per shop per month	TZS 75,000	Weighted across the Shop and Multi-branch tiers	Assumption
USD conversion	TZS 2,600 per USD	Indicative only, used to make the scale legible to non-local readers	Assumption

9.2 Revenue per Active Shop per Month

Stream	Calculation	Monthly revenue (TZS)	Share
Subscription	Blended tier average	75,000	17%
Loan facilitation	60 devices × 3,000	180,000	41%
Payment rail margin	500 repayments × 350	175,000	40%
Messaging margin	700 messages × 15	10,500	2%
Total	Per active shop per month	440,500	100%

Two observations follow from the composition. First, **subscription is the smallest stream by value**, which is the opposite of the intuition most software pricing starts from. The commercial model is transaction-led, and the subscription exists mainly to anchor the relationship and cover support. Second, facilitation and rail margin together account for roughly four

fifths of revenue and both are functions of the same underlying activity: the customer lending money and collecting it. This means the product's revenue is structurally aligned with the customer's core business, which is the strongest possible position for a B2B system.

9.3 Three-Year Adoption and Revenue Ramp

The curve below assumes a deliberately conservative adoption pattern: a pilot-led start, a regional expansion phase, and a national phase gated on operational support capacity rather than on demand. Revenue is calculated on **average** active shops during each year, not on the closing count, which is the more honest basis.

Year	Shops (closing)	Average active shops	Annual revenue (TZS)	Approx. USD	Focus
1	25	12	63,400,000	24,000	Pilot, validation, reference accounts
2	120	70	369,600,000	142,000	Regional expansion, support build-out
3	400	260	1,372,800,000	528,000	National coverage, wholesale accounts

The shape of the curve, not the magnitude, is the point. Growth is constrained by onboarding and support capacity rather than by market size, which is why the valuation of the business depends far more on how quickly a new shop can be onboarded and trained than on how many shops exist in the country.

9.4 Cost Structure

Cost category	Nature	Indicative proportion of revenue
Messaging and rail fees	Variable, per transaction and per message	Already netted in the stream figures above
Hosting, database, storage and backups	Semi-variable, scales with data and tenants	6 – 9%
Customer onboarding and training	Variable, front-loaded per new shop	5 – 8% of first-year revenue per shop
Customer support	Step-fixed, scales with installed base	4 – 7%
Aggregator and settlement fees	Variable, per settlement relationship	Included in rail margin
Engineering and maintenance	Fixed, resolved by the delivery roadmap	Not modelled here

Excluding engineering and product overhead, which is a fixed investment rather than a percentage, the operating gross margin implied by these proportions is approximately **80 to 85 percent**. That is a healthy structure for a system of this kind, and the main risk to it is support cost, which is the category most likely to be underestimated at 120 shops and beyond.

9.5 Sensitivity

Four inputs determine whether the model works. Each should be tested during the pilot before any pricing commitment is made.

- **Financed devices per shop per month.** The dominant driver. A dealership financing 25 devices rather than 60 reduces per-shop revenue by roughly 40 percent and changes the pricing conversation entirely.
- **Facilitation fee acceptance.** The fee most likely to be resisted, because it is the most visible and the most novel. It must be demonstrated to be smaller than the loss it prevents.

- **Repayment transaction volume.** Depends on how much of the loan book is collected through the integrated rails rather than in cash. If most collections stay in cash, this stream largely disappears, which is a strong argument for making the rails the default path in the counter workflow rather than an optional one.
- **Support cost per shop.** The one that most often surprises. Onboarding a shop with no digital record-keeping is a training problem before it is a software problem.

10. Go-To-Market Sequence — Tanzania

The sequence below is ordered deliberately. Each step produces evidence that the next step depends on, and the order reflects the reality that this product spreads through dealer networks rather than through advertising.

Step	Action	Evidence it produces
1	Deploy the transactional core with one or two friendly dealerships, free of charge, with full data access and weekly contact	Whether the counter workflow survives real conditions and where the offline protocol breaks
2	Add credit rails (Gaps 1 to 3) for those same shops and measure collection performance against their historical baseline	The only credible evidence that the product reduces loss, which is the entire pricing argument
3	Fix pricing on the two anchor accounts and document the outcome in writing, including what they rejected	A defensible price point supported by a real customer rather than a spreadsheet
4	Target hub-and-spoke dealer networks rather than isolated shops: winning one hub onboards its satellite shops	Cheaper acquisition and a natural referral path
5	Establish aggregator and messaging partnerships before regional expansion, negotiated on projected volume	Rail economics that improve with scale rather than degrade
6	Move to regional hubs with a train-the-trainer model: a trained operator in-region onboarding subsequent shops	Onboarding cost per shop that falls instead of rising
7	Approach wholesalers and importers with the white-label proposition once three reference accounts exist	A larger contract size with a much shorter sales cycle

Timing note: the pilot in steps 1 and 2 should not be rushed. A pilot that runs for six months and produces a documented before-and-after collection comparison is worth more commercially than ten additional installations, because it converts an unproven claim into a measured result that can be shown to every subsequent prospect.

11. Legal, Regulatory & Ethical Guardrails

The capabilities described in Section 6 are legitimate commercial functions, but three of them touch regulated or legally sensitive territory. This section identifies what must be confirmed before deployment. It is a checklist for qualified local advisers, not a substitute for them.

Area	Requirement to confirm	Risk if ignored
Device locking	Consumer protection and contract law treatment of remote device restriction; required disclosure, notice periods and release conditions; enforceability of the covenant in the customer agreement	Unenforceable terms, consumer complaints, reputational damage, potential regulatory action
Device locking — implementation route	Legal and commercial standing of the chosen mechanism: OEM-backed financing lock programme versus custom provisioning	Custom provisioning without contractual cover is the highest-exposure option
Personal data protection	Tanzania has a data protection framework requiring lawful basis, purpose limitation and security obligations. Confirm registration or notification duties, cross-border storage implications, and retention rules	Regulatory exposure and loss of enterprise customers who require compliance evidence
National identity data	Any connection to national identity records requires a formal agreement with the issuing authority. Confirm whether and how access can be obtained	Unauthorised access, and a hard block on a core feature
Fiscal receipting	Confirm obligations, thresholds and the approved integration route for electronic fiscal receipts; confirm receipt numbering and format requirements	Product cannot serve formal dealerships; retrofitting is expensive
Lending activity	Confirm what constitutes regulated lending and deposit-taking, and where interest-bearing credit requires a licensed partner rather than the platform itself	Regulatory action; invalid commercial structure
Mobile money integration	Confirm the correct contractual route: direct operator relationship versus licensed aggregator, and the data protection terms attaching to payment events	Settlement disputes, data handling breaches
Application security	Maintain the security baseline established in the technical blueprint (mobile application security verification standard), including secure storage of credentials and no secrets in the client build	Credential compromise; a breach would be existential for a system holding borrower and payment data

ETHICAL POSITION

This product should be built on the principle that **a customer in good standing never experiences the recovery mechanism**. Locking exists to make unsecured lending possible, not to punish. If the release path on payment is slower or less reliable than the lock path on default, the design is wrong regardless of what the contract permits.

12. Risk Register for the New Capabilities

Risk	Likelihood	Impact	Mitigation
Device locking produces customer disputes and complaints	Medium	High	Full disclosure in the agreement; written notice before locking; instant automatic release on payment; a documented dispute path with a named owner
Recovery mechanism can be bypassed on unlocked devices	Medium	High	Provision at point of sale; dual-IMEI trapping (Gap 5); lock event ledger; periodic integrity checks on financed devices
Payment webhooks arrive duplicated or out of order	High	Medium	Idempotent matching on a payment reference; an unmatched-event reconciliation queue; never infer a payment from an amount alone
Identity data mishandled or over-retained	Medium	High	Purpose limitation and consent recorded at capture; documented retention periods; encryption at rest; access restricted by role with RLS
Guarantor consent is taken informally and is unenforceable	Medium	Medium	Structured consent capture with a recorded acknowledgement and verified contact details for the guarantor
Agent consignments quietly become losses	Medium	Medium	Settlement deadlines with automated alerts; outstanding-device discrepancy reporting; agent performance visibility before extending more stock
Credit scoring rules encode bias or are unexplainable	Low	High	Keep the v1 rule explicit and human-readable; record the factors used for each decision; review outcomes across branches periodically
Support cost per shop exceeds the pricing assumption	High	Medium	Train-the-trainer model in-region; measure onboarding hours per shop from the first pilot and treat it as a pricing input
Local network conditions make rails unreliable at the counter	Medium	Medium	Offline-first queue for collection intents; a documented fallback procedure with the same audit trail as the online path
Locking capability misused by staff or by a reseller	Low	High	Every lock transition requires a recorded authorisation with an actor identity; lock actions reported in the daily closing and in management reports

13. Implementation Addendum — Schema Hooks

The following tables are additive. They extend the schema described in the technical blueprint without altering the twelve existing tables, which means they can be deployed in a later migration without disrupting the transactional core. Every table carries a client operation identifier where it participates in offline capture, and every table is subject to row-level security consistent with the existing model.

The purpose of listing them here is sequencing rather than completeness: the shape of the data model determines what the product can later do, and these decisions are cheap now and expensive after launch.

13.1 Device Recovery (Gap 1 and Gap 5)

```
CREATE TYPE lock_state AS ENUM (
'unlocked', 'financed', 'grace', 'locked',
'released', 'recovery_initiated', 'recovered', 'written_off'
);

CREATE TABLE IF NOT EXISTS device_locks (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
phone_id UUID NOT NULL REFERENCES phones(id),
loan_id UUID REFERENCES loans(id),
imei_primary VARCHAR(50) NOT NULL,
imei_secondary VARCHAR(50),
state lock_state NOT NULL DEFAULT 'unlocked',
provider VARCHAR(100),
provider_device_ref VARCHAR(255),
grace_until TIMESTAMP WITH TIME ZONE,
locked_at TIMESTAMP WITH TIME ZONE,
released_at TIMESTAMP WITH TIME ZONE,
created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

CREATE TABLE IF NOT EXISTS lock_events (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
device_lock_id UUID NOT NULL REFERENCES device_locks(id) ON DELETE CASCADE,
from_state lock_state,
to_state lock_state NOT NULL,
reason TEXT NOT NULL,
actor_id UUID REFERENCES employees(id),
evidence_ref VARCHAR(255),
occurred_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

CREATE INDEX idx_locks_phone ON device_locks(phone_id);
CREATE INDEX idx_locks_state ON device_locks(state);
CREATE INDEX idx_lock_events_d ON lock_events(device_lock_id, occurred_at DESC);
```

13.2 Collection Rails (Gap 2)

```
CREATE TABLE IF NOT EXISTS payment_intents (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
loan_id UUID NOT NULL REFERENCES loans(id),
schedule_id UUID REFERENCES loan_schedules(id),
expected_amount DECIMAL(12,2) NOT NULL,
payer_msisdn VARCHAR(20) NOT NULL,
provider VARCHAR(50) NOT NULL,
```

```
client_operation_id UUID UNIQUE NOT NULL,
status VARCHAR(50) DEFAULT 'pending',
provider_ref VARCHAR(255),
created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
resolved_at TIMESTAMP WITH TIME ZONE
);

CREATE TABLE IF NOT EXISTS payment_events (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
provider VARCHAR(50) NOT NULL,
provider_txn_id VARCHAR(255) UNIQUE NOT NULL,
payer_msisdn VARCHAR(20),
amount DECIMAL(12,2) NOT NULL,
intent_id UUID REFERENCES payment_intents(id),
matched_loan_id UUID REFERENCES loans(id),
match_status VARCHAR(50) DEFAULT 'unmatched',
raw_payload JSONB NOT NULL,
received_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

CREATE INDEX idx_events_match ON payment_events(match_status);
CREATE INDEX idx_events_payer ON payment_events(payer_msisdn);
CREATE INDEX idx_events_loan ON payment_events(matched_loan_id);
```

13.3 Identity, Guarantors and Credit (Gap 3 and Gap 7)

```
CREATE TABLE IF NOT EXISTS kyc_verifications (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
customer_id UUID NOT NULL REFERENCES customers(id),
id_document_type VARCHAR(50) NOT NULL,
id_document_number VARCHAR(100) NOT NULL,
id_document_photo TEXT,
selfie_photo TEXT,
verification_method VARCHAR(50) NOT NULL,
verification_status VARCHAR(50) DEFAULT 'pending',
consent_recorded_at TIMESTAMP WITH TIME ZONE,
purpose TEXT NOT NULL,
verified_by UUID REFERENCES employees(id),
verified_at TIMESTAMP WITH TIME ZONE,
created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

CREATE TABLE IF NOT EXISTS guarantors (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
loan_id UUID NOT NULL REFERENCES loans(id) ON DELETE CASCADE,
full_name VARCHAR(255) NOT NULL,
msisdn VARCHAR(20) NOT NULL,
id_document_no VARCHAR(100),
relationship VARCHAR(100),
consent_at TIMESTAMP WITH TIME ZONE NOT NULL,
liability_note TEXT,
created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

CREATE TABLE IF NOT EXISTS credit_assessments (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
customer_id UUID NOT NULL REFERENCES customers(id),
loan_id UUID REFERENCES loans(id),
score INTEGER NOT NULL,
band VARCHAR(20) NOT NULL,
```

```
factors JSONB NOT NULL,
decision VARCHAR(50) NOT NULL,
decided_by UUID REFERENCES employees(id),
override_note TEXT,
assessed_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

CREATE TABLE IF NOT EXISTS collections_actions (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
loan_id UUID NOT NULL REFERENCES loans(id),
ladder_state VARCHAR(50) NOT NULL,
action_type VARCHAR(50) NOT NULL,
channel VARCHAR(50),
outcome TEXT,
customer_said TEXT,
actor_id UUID REFERENCES employees(id),
occurred_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

13.4 Agents, Consignments and Blacklist (Gap 4 and Gap 5)

```
CREATE TABLE IF NOT EXISTS agents (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
full_name VARCHAR(255) NOT NULL,
msisdn VARCHAR(20) NOT NULL,
id_document_no VARCHAR(100),
branch_id UUID REFERENCES branches(id),
terms TEXT,
status VARCHAR(50) DEFAULT 'active',
created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

CREATE TABLE IF NOT EXISTS consignments (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
agent_id UUID NOT NULL REFERENCES agents(id),
from_branch_id UUID REFERENCES branches(id),
issued_at TIMESTAMP WITH TIME ZONE NOT NULL,
settlement_due_at TIMESTAMP WITH TIME ZONE NOT NULL,
status VARCHAR(50) DEFAULT 'open',
settled_at TIMESTAMP WITH TIME ZONE,
client_operation_id UUID UNIQUE NOT NULL
);

CREATE TABLE IF NOT EXISTS consignment_lines (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
consignment_id UUID NOT NULL REFERENCES consignments(id) ON DELETE CASCADE,
phone_id UUID NOT NULL REFERENCES phones(id),
imei_primary VARCHAR(50) NOT NULL,
outcome VARCHAR(50) DEFAULT 'outstanding',
settled_amount DECIMAL(12,2),
resolved_at TIMESTAMP WITH TIME ZONE,
UNIQUE (consignment_id, phone_id)
);

CREATE TABLE IF NOT EXISTS blacklist_registry (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
imei VARCHAR(50) UNIQUE NOT NULL,
reason VARCHAR(100) NOT NULL,
source VARCHAR(100) NOT NULL,
note TEXT,
```

```
flagged_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
cleared_at TIMESTAMP WITH TIME ZONE
);

CREATE TABLE IF NOT EXISTS trade_in_valuations (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
brand VARCHAR(100) NOT NULL,
model VARCHAR(150) NOT NULL,
storage_gb INTEGER,
condition VARCHAR(50) NOT NULL,
offer_amount DECIMAL(12,2) NOT NULL,
effective_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
UNIQUE (brand, model, storage_gb, condition)
);
```

13.5 Fiscal Receipting (Gap 6)

```
CREATE TABLE IF NOT EXISTS fiscal_receipts (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
sale_id UUID NOT NULL REFERENCES sales(id),
receipt_number VARCHAR(100) NOT NULL UNIQUE,
fiscal_device_no VARCHAR(100),
gross_amount DECIMAL(12,2) NOT NULL,
tax_amount DECIMAL(12,2),
currency VARCHAR(10) DEFAULT 'TZS',
issued_at TIMESTAMP WITH TIME ZONE NOT NULL,
transmission_state VARCHAR(50) DEFAULT 'pending',
acknowledgement JSONB,
created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

CREATE INDEX idx_fiscal_sale ON fiscal_receipts(sale_id);
CREATE INDEX idx_fiscal_state ON fiscal_receipts(transmission_state);
```

13.6 Sequencing Note

Deploying all of the above in the first migration is not necessary. What matters is that **the identifiers and states are decided before the transactional core is populated with real data**, because the two hardest retrofits are the dual-IMEI pairing on existing device records and the borrower identity linkage on existing loans. The lock state machine, consignment custody state and fiscal receipt reference can be added later with far less friction.

14. Monetisation Roadmap by Delivery Phase

Revenue is switched on progressively, gated on capability rather than on calendar. Charging for a capability before it is demonstrably working damages the reference value of the early customers, and early customers in this market are the hardest to replace.

Phase	Capability delivered	Commercial activation	Gate condition
0	Repository layer, real Supabase connection, credentials moved out of the client	None — pilot shops run free	Live data flowing in at least one real shop
1	Transactional core, inventory movements, transfers, daily closing, dual-IMEI registration	Subscription tier priced but not charged	Two weeks of uninterrupted daily closing with no data loss
2	Loan schedules, repayment rails, webhook reconciliation, KYC capture	First paid subscription; messaging margin live	Collection comparison against the shop's own baseline documented
3	Device locking via the supported route, lock event ledger, collections ladder	Facilitation fee activated	Legal review completed and agreement wording approved
4	Agents, consignments, blacklist registry, trade-in valuation	Multi-branch tier and expansion pricing	Agent settlement workflow operating in at least one dealer
5	Fiscal receipting, white-label packaging, guarded analytics	Wholesale and white-label contracts	Reference accounts available and compliance terms settled

15. Recommendations & Decision Points

Eight decisions are required before the next build phase can be scoped. Each is stated as a question with a recommended answer, because leaving them open is itself a decision with consequences.

#	Decision required	Recommendation
1	Is this product positioned as a stock system or as a credit control system?	Credit control, with inventory as its foundation. This determines the pricing model and the roadmap order.
2	Which device locking route is adopted?	The OEM-backed financing lock route. Legal review before any implementation work begins.
3	Which payment aggregator is selected?	Selection driven by webhook reliability and settlement terms, not by headline rate. Non-trivial to change later.
4	Is identity verification manual or integrated in v1?	Manual document and selfie capture in v1, with the structure in place for an authority integration later.
5	How much of the loan book must run through integrated rails?	Set an explicit target as a product requirement, since this stream disappears if collections stay in cash.
6	Which dealership is the pilot, and what access is agreed?	One friendly, higher-volume dealership with an existing loan book and a willingness to share baseline collection data.
7	Is a licensed lending partner required?	Confirm with qualified legal advice before any interest-bearing structure is offered on the platform.
8	What is the onboarding cost per shop, measured rather than assumed?	Instrument it from the first installation. It is the single largest risk to the margin model.

FIRST FOURTEEN DAYS

Three actions: (1) shortlist two payment aggregators and request webhook documentation and settlement terms; (2) obtain initial legal guidance on the device locking covenant and consumer disclosure; (3) agree the pilot dealership and ask it for twelve months of collection data. None of the three requires engineering capacity, and together they de-risk the two largest assumptions in this document.

16. Muhtasari wa Kiswahili

Sehemu hii ni muhtasari mfupi kwa Kiswahili, kwa ajili ya mazungumzo na wadau wa ndani na wateja wa awali.

16.1 Hitimisho Kuu

Mfumo ulioandikwa kwenye technical specification ni mzuri kama msingi wa kurekodi mauzo na kufuatilia bidhaa. Lakini biashara ya simu Tanzania haipati faida kubwa kwenye kuuza simu yenyewe — faida halisi ipo kwenye kukopesha, kukusanya marejesho, na kupunguza hasara ya wateja wasiolipa. Kwa hivyo mfumo unapaswa ujengwe kama mfumo wa kudhibiti mikopo, na hesabu ya bidhaa iwe msingi wake tu.

16.2 Mapungufu Saba Muhimu

- **Kufunga simu (device locking) na urejeshaji.** Bila njia ya kurejesha, mkopo usiolipwa ni hasara kamili. Hii ni hatua yenye manufaa makubwa zaidi.
- **Marejesho ya moja kwa moja kwa njia ya mobile money.** Kukusanya kwa mkono ni polepole, kunagharimu muda, na kunarahisisha udanganyifu.
- **Utambulisho wa mkopaji, KYC na wadhamini.** Bila utambulisho uliohakikiwa, haiwezekani kupima hatari wala kusimamia mikopo kwa usalama.
- **Mawakala na bidhaa za kuaminiana (consignment).** Sehemu kubwa ya simu zinapita kwa watu wa kati; mfumo usiowajumuisha utahesabu vibaya.
- **IMEI mbili, trade-in na orodha ya simu zilizoibwa.** Simu nyingi zina IMEI mbili; kushika moja tu huacha njia ya kuzunguka.
- **Risiti za kodi (EFD/VFD).** Bila hivi, maduka makubwa rasmi hayawezi kutumia mfumo kama kitabu chao kikuu.
- **Maamuzi ya mikopo na usimamizi wa madeni.** Mpangilio wa hatua za kufuatilia deni unapaswa kuwa sehemu ya mfumo, si ripoti tu.

16.3 Njia za Mapato

- Usajili wa mwezi kwa kila tawi: TZS 40,000 hadi 500,000 kutegemea kiwango.
- Ada kwa kila simu inayotolewa kwa mkopo kupitia mfumo: TZS 2,000 hadi 5,000.
- Faida ndogo kwa kila marejesho yanayopita kwenye njia za malipo: TZS 300 hadi 500.
- Faida kwenye SMS na taarifa za ukumbusho: TZS 25 hadi 35 kwa ujumbe (gharama TZS 12 hadi 15).
- White-label kwa wauzaji wa jumla: ada ya kuanzisha TZS milioni 3 hadi 7 pamoja na malipo ya mwezi.
- Uchambuzi wa hatari kwa muunganisho wa data (bila taarifa za kibinafsi): hatua ya mwisho kabisa.

16.4 Mipaka ya Kisheria

Kufunga simu kunakubalika tu ikiwa kumeandikwa kwenye mkataba, kumeelezwa kwa mteja, na kufunguliwa mara moja baada ya malipo. Ukusanyaji wa taarifa za kibinafsi unapaswa kufuata sheria ya ulinzi wa taarifa za kibinafsi, na uhusiano na mamlaka ya vitambulisho unahitaji makubaliano rasmi. Ushauri wa kitaalamu wa kisheria unahitajika kabla ya kuanza.

Appendix A — Terminology

Term	Meaning
Aggregator	A service that connects a business to multiple mobile money operators through a single interface and settlement relationship.
Arrears ladder	The defined sequence of states and actions a loan passes through as it becomes overdue, each with a trigger, an owner and a required record.
Client operation identifier	A value generated on the client for each transaction so the server can recognise a replayed submission instead of posting it twice.
Consignment	Stock issued to an agent or reseller that remains the dealership's property until sold or returned.
Custody	Which party is physically holding and responsible for a device at a given moment.
Dual-IMEI	A handset carrying two distinct IMEI values, one per SIM slot. Both must be recorded and constrained for uniqueness.
IMEI	The unique identifier of a mobile handset modem. Used in this system as the device's identity key.
Idempotent	An operation that produces the same result whether it is executed once or several times, which is what makes retries safe.
Outbox	A local queue of operations waiting to be sent to the server, holding transactions created while offline.
Row-level security	A database feature restricting which rows a given user may read or write, based on their identity and role.
Trade-in	A used handset accepted as part payment, which requires valuation, provenance checking and intake registration.

Sign-Off & Next Review

This document is a discussion draft for product and commercial review. It does not amend the technical specification, which remains the controlling document for implementation.

Role	Name	Responsibility	Status
Lead Developer	Godfrey Shora	Implementation lead, schema extensions, collection rails, identity & consignment data model	Approved
Architecture & Strategy	David Erick Mwijage	Product architecture, credit and recovery design, commercial model	Approved
Commercial	To be confirmed	Pricing validation, pilot agreements, partner negotiation	Pending
Legal review	To be confirmed	Device locking covenant, data protection, fiscal receipting, lending structure	Required before Phase 3

NEXT ACTION

Confirm the pilot dealership and request twelve months of collection data. That single dataset converts the largest assumption in Section 9 into a measured input, and it costs nothing to obtain.